

Operations — Instrumentation · Maintenance · Change Management

How the pipeline is observed, kept healthy, and changed safely.

1 · Instrumentation plan — what's tracked, and how I'd know it's working

- **Per run (built):** tier distribution + rep load, written to `output/` and compared against the prior run (`memory/runs/`) by the decision brief — deltas, enrichment trust score (% live-signal vs. fallback), and rules-triggered recommendations.
- **Pipeline health (Trigger.dev):** per-stage success/retries/duration; a failed enrichment degrades one lead's confidence, never blocks the run.
- **LLM (Langfuse):** cost, tokens, latency per call; groundedness eval on every battle card — a card citing an unverified number fails before a rep sees it.

Alert thresholds (Slack, ops channel)

Alert	Meaning
Any unrouted lead	Invariant breach — run fails loudly
Tier skew: 0 hot, or hot >40%	Rubric miscalibrated
Rep imbalance: >50% of book on one rep	Routing rules need a territory rebalance
Enrichment trust <50%	Don't trust the vertical mix until sources improve

How I'd know it's actually working

Day one: 30/30 exit with rep + score + human-readable reason, zero unrouted, re-runs idempotent (`npm run check`) — offline invariant tests for scoring, routing, suppression, ownership).

Real signal: reply/meeting rate **by tier**. If hot doesn't outperform warm, the rubric is decoration. Rep accept/decline decisions (with reasons) are the training data that retunes weights — the loop learns in data, not vibes.

2 · Maintenance plan — what needs regular attention

Cadence	Task	Trigger to act early
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Weekly	Review Langfuse spend + eval pass-rate; triage any failed runs (per-stage retry logs)	Eval pass-rate drops, cost spike
Weekly	Sync <code>data/socure-customer-list.csv</code> from CRM — suppression is only as good as this list	New closed-won logo
Monthly	Vendor coverage check: Bitscale/nRev hit-rate vs. the DNS/RDAP + symbolic floor	Enrichment trust score trending down
Monthly	Reconcile ownership ledger vs. CRM account owners	Rep change, territory change
Quarterly	Refresh grounding corpus: new Socure case studies enter the allowlist only after adversarial verification ; retired claims join the blocklist	New press release cited in a card draft
Quarterly	Rubric recalibration: re-run the reverse-ICP on the current customer base; retune weights from the rep accept/decline ledger	Hot-tier reply rate $\leq$ warm-tier
As released	Pin bumps: Trigger.dev SDK, Anthropic model version (Haiku deprecations), Langfuse	Deprecation notices

Failure runbook: every stage retries independently (Trigger.dev); an invariant breach (unrouted lead, contract violation) fails the run loudly by design — fix config or data, re-run; idempotency guarantees no duplicate rows, cards, or events.

3 · Change management plan — how changes ship safely

- **All decision logic is config, so every change is a diff.** The ICP, territories, and confidence ceilings live in `config/*.json` (the rubric carries a `version` field and inline evidence citations). No weights or thresholds hide in code or prompts — a change is a reviewable one-file PR, and rollback is `git revert` + re-run (idempotent, so regeneration is safe).
- **Shadow before promote.** A rubric or routing change is re-run against the current book before it goes live; the decision brief diffs the result vs. the prior snapshot (`memory/runs/`) — tier migrations and rep-load shifts are visible *before* any rep's book actually moves. Promote only when the diff is intended.
- **Evidence gate on content changes.** New proof points, competitive claims, or persona pains enter `docs/grounding/` only through verification; the battle-card eval enforces it downstream, so an unverified claim can't ship even if it sneaks into a prompt.
- **Prompt changes are versioned and eval'd.** Prompts live in `prompts/` (files, so they diff like code); Langfuse traces which version produced which output, and the groundedness eval must hold before a new prompt version becomes the default.

- **Human gates only loosen by evidence — and some never do.** Autonomy changes follow the graduated-trust ladder (per-category confidence $\geq \sim 90\%$ over ≥ 20 decided items to batch review; rolling-window demotion below $\sim 80\%$). Battle cards are outward-facing → permanent per-item rep review, regardless of confidence.
- **People side:** reps get a Slack note when territories or the rubric change (their book shifts); the decision brief's next run quantifies exactly what moved and why (variance attributed by rubric dimension), so the change explains itself.

Socure Growth Engineering Exercise · source of truth: [docs/OPERATIONS.md](#)